VALIDATION PLAN
OPERATIONAL QUALIFICATION
MSA MODULE

JR Validated Environment — MSA Module v1.0

Field	Value
Document Number	JR-VP-MSA-001
Title	Validation Plan — Operational Qualification, MSA Module
System	JR Validated Environment — MSA Module
Module Version	1.0
Document Version	1.0
Status	Draft
Effective Date	2026-03-18
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-18	Joep Rous	Initial release. Covers all 5 MSA scripts in module v1.0.

Approval Signatures

Role	Name	Signature	Date
Author			
Reviewer			
Approver (QA)			

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the MSA (Measurement System Analysis) module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each MSA script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the MSA module.

2. Scope

2.1 In Scope

- All 5 R scripts in the repos/msa/R/ directory
- Correct computation of outputs for valid inputs (Kappa, variance components, %GRR, Cg/Cgk, bias/linearity)
- Correct rejection of invalid inputs with informative error messages
- Correct file I/O behaviour (CSV input, PNG output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup
- Balanced-design validation for crossed and nested Gauge R&R

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Demo scripts jrc_msa_R_hello.R and jrc_msa_py_hello.py
- Performance qualification (PQ) — end-to-end studies using production data are conducted separately under individual design verification protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, grid) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Kappa, ANOVA, Weibull) — method references are cited in Section 3

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 11	Electronic Records; Electronic Signatures

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002

FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Standards and Methods

Reference	Title
AIAG MSA, 4th Ed.	Measurement Systems Analysis Reference Manual, Automotive Industry Action Group, 2010
ISO 22514-7:2021	Statistical methods in process management — Capability and performance — Part 7: Capability of measurement processes
ISO 13485:2016	Medical devices — Quality management systems — Requirements for regulatory purposes
Fleiss (1971)	Fleiss JL. Measuring nominal scale agreement among many raters. Psychological Bulletin, 76(5), 378–382.
Cohen (1960)	Cohen J. A coefficient of agreement for nominal scales. Educational and Psychological Measurement, 20(1), 37–46.

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment v1.0.0
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_20260311_205146.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite v1.1.0

4. Definitions and Abbreviations

Term	Definition
MSA	Measurement System Analysis — a body of techniques for characterising the ability of a measurement system to produce accurate and repeatable results
GRR	Gauge Repeatability and Reproducibility — a study to quantify the variation introduced by the measurement system relative to total process variation
%GRR	Percentage of total study variation attributable to the measurement system; < 10% Acceptable, 10–30% Marginal, > 30% Unacceptable
EV	Equipment Variation (Repeatability) — variation due to the gauge itself
AV	Appraiser Variation (Reproducibility) — variation due to operators
PV	Part Variation — true part-to-part variation
ndc	Number of Distinct Categories — how many groups the gauge can reliably distinguish; ≥ 5 is required
Cg	Gauge Capability index (Type 1 study) — ratio of tolerance to measurement spread; ≥ 1.33 required
Cgk	Gauge Capability index adjusted for bias (Type 1 study); ≥ 1.33 required
Kappa	Cohen's/Fleiss' Kappa — a statistic measuring inter-rater agreement beyond what is expected by chance; 0 = chance, 1 = perfect agreement
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
UR	User Requirement
TC	Test Case
jrrun	The validated zsh entry-point script that sets the controlled environment before executing any community script
CSV	Comma-separated values input file
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. System Description

5.1 Module Overview

The MSA module is an extension of the JR Validated Environment providing five R scripts for Measurement System Analysis. Scripts are installed under repos/msa/ and are invoked via wrapper commands (e.g., jrc_msa_gauge_rr). The module shares the core environment infrastructure (jrrun, renv library, integrity checking, logging) with the community script suite.

5.2 Scripts Under Test

Script	Method	Input Columns	Key Outputs
jrc_msa_gauge_rr	Two-way ANOVA (crossed)	part, operator, value	%GRR, ndc, variance components, 4-panel PNG
jrc_msa_nested_grr	One-way nested ANOVA	operator, part, replicate, value	%GRR, variance components, 2-panel PNG
jrc_msa_linearity_bias	Linear regression	part, reference, value	Slope, intercept, %Linearity, per-part bias, 2-panel PNG
jrc_msa_type1	Type 1 gauge study	value (optional: id)	Cg, Cgk, bias t-test, 2-panel PNG
jrc_msa_attribute	Cohen's / Fleiss' Kappa	part, appraiser, trial, rating (optional: reference)	Within/between Kappa, % agreement, 2-panel PNG

5.3 Intended Use

MSA scripts are intended for use by engineers and scientists qualifying measurement systems used in medical device design verification. Intended uses include:

- Quantifying the repeatability and reproducibility of a measurement gauge before using it in a design verification study
- Determining whether a gauge is capable of resolving real part-to-part differences
- Assessing gauge linearity and bias across its operating range
- Evaluating the capability of a gauge to measure within a specification tolerance
- Qualifying attribute inspection systems (pass/fail, categorical)
- Supporting measurement system qualification records in the Design History File (DHF)

5.4 Hardware and Software Environment

Component	Specification
Operating system	macOS (arm64 / aarch64-apple-darwin)
R version	4.5 (as specified in admin/r_version.txt)
R packages	ggplot2, grid (versions as locked in admin/renv.lock)
Entry point	bin/jrrun
Module location	repos/msa/
Test runner	pytest in ~/.venvs/MyProject_oq/ (reused from core OQ venv)

6. Risk Assessment and Validation Level

6.1 Risk Classification

MSA scripts are used to determine whether a measurement system is fit for purpose in a design verification study. Incorrect results could lead to accepting a measurement system with unacceptable gauge error, leading to unreliable verification data, incorrect pass/fail conclusions, and potentially unsafe medical devices reaching

the market. This represents a high-consequence failure mode. Per the FDA General Principles of Software Validation (GPSV 2002), the level of validation effort is commensurate with this risk.

6.2 Risk Factors

Risk Factor	Assessment	Consequence
Incorrect variance component calculation (%GRR)	Medium probability	High: accept a poor gauge for use in DHF testing
Incorrect Kappa calculation	Medium probability	High: accept an unreliable inspection system
Design balance validation bypassed	Low probability	High: biased estimates from unbalanced data
Silent failure (non-zero exit, no message)	Low probability	Medium: user unaware of failure
Script runs outside jrrun (wrong library)	Low probability	High: wrong package versions, undetected results

Strategy: Full OQ testing of all 5 scripts, covering both primary (happy-path) execution paths and boundary/error-handling paths. Numerical outputs are verified against independently determined expected ranges using known test datasets.

7. Roles and Responsibilities

Role	Responsibility
Author	Writes and maintains this Validation Plan and the MSA module scripts
Reviewer	Reviews test specifications for completeness, accuracy, and regulatory adequacy
QA Approver	Approves this plan prior to OQ execution
Test Executor	Executes the OQ test suite (admin_msa_oq) and documents results
QA Witness	Reviews execution evidence and signs the OQ execution report

The Test Executor and Author may be the same person. The QA Approver must be independent of the Author for this document.

8. Validation Approach

8.1 Overall Strategy

The MSA module is validated as an extension of the core JR Validated Environment using the same IQ/OQ/PQ model. The core environment IQ is documented in JR-IQ-001. This document covers the OQ for the MSA module only.

Phase	Description	Status
IQ	Core environment: R version, packages, jrrun, integrity checking. MSA module scripts are included in the project_integrity.sha256 file.	Completed — JR-IQ-001
OQ (this document)	Verifies correct operation of each MSA script across the range of anticipated inputs	Planned
PQ	End-to-end performance with real measurement data; executed under individual design verification protocols	Out of scope here

8.2 OQ Execution

OQ tests are executed by the automated test runner repos/msa/admin_msa_oq, which:

1. Activates the OQ Python virtual environment (~/.venvs/MyProject_oq/)
2. Invokes pytest against all test_*.py files in repos/msa/oq/
3. For each test case, calls the MSA script as a subprocess via jrrun
4. Captures exit code, stdout, and stderr
5. Evaluates pass/fail against the criteria defined in Section 13
6. Writes a timestamped qualification report to ~/.jrscript/MyProject/validation/

All test cases must pass for the OQ to be considered successful. A single failing test case constitutes an OQ failure and must be resolved before the module is released to users.

8.3 Sequence of Execution

7. Confirm test data files in repos/msa/oq/data/ are present and committed
8. Run admin/admin_create_hash to regenerate project integrity file
9. Obtain QA approval of this plan
10. Execute repos/msa/admin_msa_oq on the target machine
11. Review the qualification report
12. Document any deviations
13. Obtain QA sign-off on the execution evidence

9. User Requirements

The following user requirements define what the MSA module must do from the user's perspective. Each requirement is traceable to one or more OQ test cases in the Requirements Traceability Matrix (Section 14).

9.1 Gauge R&R (Crossed)

ID	Requirement
UR-MSA-001	The system shall perform a Gauge R&R analysis using the two-way ANOVA method (crossed design) on a balanced part/operator/value dataset, computing: ANOVA table, variance components, repeatability (EV), reproducibility (AV), %GRR vs study variation, %GRR vs tolerance (when supplied), and number of distinct categories (ndc).
UR-MSA-002	The system shall classify the measurement system as ACCEPTABLE (%GRR < 10%), MARGINAL (10% ≤ %GRR < 30%), or UNACCEPTABLE (%GRR ≥ 30%) and display the verdict.
UR-MSA-003	The system shall save a four-panel PNG (components of variation, by-part, by-operator, interaction plot) to ~/Downloads/.
UR-MSA-004	The system shall reject datasets that are unbalanced, have fewer than 2 operators or parts, or are missing required columns, with an informative error message.

9.2 Nested Gauge R&R (Destructive)

ID	Requirement
UR-MSA-005	The system shall perform a Nested Gauge R&R analysis on a balanced nested dataset (parts nested within operators), computing: ANOVA table, variance components, repeatability (EV), reproducibility (AV), part-within-operator variation, %GRR vs study variation, and %GRR vs tolerance (when supplied).
UR-MSA-006	The system shall classify the nested measurement system as ACCEPTABLE, MARGINAL, or UNACCEPTABLE using the same %GRR thresholds as UR-MSA-002.
UR-MSA-007	The system shall save a two-panel PNG (components of variation, measurements by operator) to ~/Downloads/.
UR-MSA-008	The system shall reject unbalanced nested datasets or datasets missing required columns, with an informative error message.

9.3 Linearity and Bias

ID	Requirement
UR-MSA-009	The system shall perform a linearity and bias analysis by regressing observed bias (measured value – reference) against reference value, reporting: regression slope and intercept with p-values, R^2, %Linearity, and per-part bias with 95% CI and significance flag.
UR-MSA-010	The system shall classify the gauge as having ACCEPTABLE or UNACCEPTABLE linearity and bias based on slope significance ($p < 0.05$) and per-part bias significance.
UR-MSA-011	The system shall save a two-panel PNG (linearity regression plot with 95% CI band, per-part bias bar chart) to ~/Downloads/.
UR-MSA-012	The system shall reject datasets with fewer than 2 distinct reference values, inconsistent reference values per part, or missing required columns.

9.4 Type 1 Gauge Study

ID	Requirement
UR-MSA-013	The system shall perform a Type 1 gauge capability study on repeated measurements of a single reference part, computing: $Cg = 0.2 \times T / (6\sigma)$, $Cgk = (0.1 \times T - \lvert bias \rvert) / (3\sigma)$, bias, SD, and a t-test for bias significance.
UR-MSA-014	The system shall classify the gauge as CAPABLE ($Cg \geq 1.33$ and $Cgk \geq 1.33$) or INCAPABLE, and indicate when $Cgk < Cg$ (bias present).
UR-MSA-015	The system shall save a two-panel PNG (run chart with acceptance band, histogram with normal curve) to ~/Downloads/.
UR-MSA-016	The system shall require both --reference and --tolerance arguments and reject datasets with fewer than 10 measurements.

9.5 Attribute Agreement Analysis

ID	Requirement
UR-MSA-017	The system shall perform an attribute agreement analysis on a balanced part/appraiser/trial/rating dataset, computing: within-appraiser % agreement and Cohen's Kappa for each appraiser, and between-appraiser Fleiss' Kappa.
UR-MSA-018	When a reference column is present, the system shall additionally compute each appraiser's Kappa versus the reference.
UR-MSA-019	The system shall classify each Kappa as ACCEPTABLE (≥ 0.9), MARGINAL (0.7–0.9), or UNACCEPTABLE (< 0.7) and display the verdict.
UR-MSA-020	The system shall save a two-panel PNG (% agreement bar chart, Kappa chart) to ~/Downloads/.
UR-MSA-021	The system shall reject datasets that are unbalanced, have fewer than 2 appraisers, are missing required columns, or contain only one rating category.

9.6 Common Requirements

ID	Requirement
UR-MSA-022	Each MSA script shall verify that RENV_PATHS_ROOT is set at startup and terminate with a non-zero exit code and descriptive error if it is not set, preventing execution outside the jrrun validated environment.
UR-MSA-023	Each MSA script shall terminate with a non-zero exit code and display a usage message when invoked with no arguments.
UR-MSA-024	Each MSA script shall terminate with a non-zero exit code and display an informative error message when the input file is not found.

10. Acceptance Criteria

The OQ is considered successful when all of the following criteria are met:

- All 53 automated test cases defined in Section 13 pass with exit code 0 from pytest
- The qualification report generated by admin_msa_oq records 0 failures and 0 errors
- The qualification report is reviewed and signed by the QA Witness
- Any deviations are documented and dispositioned per Section 15

11. Test Data

All test data files are synthetic CSV files committed to repos/msa/oq/data/. They are version-controlled and included in the project integrity check. No real patient or production data is used.

File	Used By	Purpose
gauge_rr_balanced.csv	TC-MSA-GRR-001..004	10 parts × 3 operators × 3 reps, low noise
gauge_rr_missing_col.csv	TC-MSA-GRR-007	Missing 'operator' column
gauge_rr_unbalanced.csv	TC-MSA-GRR-008	Unequal replicates per cell
gauge_rr_one_operator.csv	TC-MSA-GRR-009	Single operator only
nested_grr_good.csv	TC-MSA-NGR-001..005	5 parts × 3 operators × 2 reps, low noise
nested_grr_poor.csv	TC-MSA-NGR-003	High within-operator variation
nested_grr_missing_col.csv	TC-MSA-NGR-008	Missing 'replicate' column
nested_grr_one_operator.csv	TC-MSA-NGR-009	Single operator only
nested_grr_unbalanced.csv	TC-MSA-NGR-010	Unequal parts per operator
linearity_bias_good.csv	TC-MSA-LB-001..004	5 reference levels × 3 reps, linear bias
linearity_bias_missing_col.csv	TC-MSA-LB-007	Missing 'reference' column
linearity_bias_one_part.csv	TC-MSA-LB-008	Single reference level
linearity_bias_inconsistent_ref.csv	TC-MSA-LB-009	Same part, different reference values
type1_good.csv	TC-MSA-T1-001..004	25 measurements of reference part, low bias
type1_biased.csv	TC-MSA-T1-003	25 measurements with systematic bias
type1_missing_value_col.csv	TC-MSA-T1-009	Missing 'value' column
type1_too_few.csv	TC-MSA-T1-010	Only 5 measurements (< 10 required)
attribute_with_ref.csv	TC-MSA-ATT-001,003..005	3 appraisers × 10 parts × 2 trials + reference
attribute_no_ref.csv	TC-MSA-ATT-002	Same without reference column
attribute_missing_col.csv	TC-MSA-ATT-008	Missing 'trial' column
attribute_one_appraiser.csv	TC-MSA-ATT-009	Single appraiser only
attribute_unbalanced.csv	TC-MSA-ATT-010	Unequal trials per appraiser-part cell

12. Requirements Traceability Summary

Requirement	Test Cases
UR-MSA-001	TC-MSA-GRR-001, TC-MSA-GRR-002, TC-MSA-GRR-003
UR-MSA-002	TC-MSA-GRR-002
UR-MSA-003	TC-MSA-GRR-004
UR-MSA-004	TC-MSA-GRR-007, TC-MSA-GRR-008, TC-MSA-GRR-009
UR-MSA-005	TC-MSA-NGR-001, TC-MSA-NGR-002, TC-MSA-NGR-004
UR-MSA-006	TC-MSA-NGR-002, TC-MSA-NGR-003
UR-MSA-007	TC-MSA-NGR-005
UR-MSA-008	TC-MSA-NGR-008, TC-MSA-NGR-009, TC-MSA-NGR-010
UR-MSA-009	TC-MSA-LB-001, TC-MSA-LB-002, TC-MSA-LB-003
UR-MSA-010	TC-MSA-LB-002, TC-MSA-LB-004
UR-MSA-011	TC-MSA-LB-005
UR-MSA-012	TC-MSA-LB-007, TC-MSA-LB-008, TC-MSA-LB-009

UR-MSA-013	TC-MSA-T1-001, TC-MSA-T1-002, TC-MSA-T1-003
UR-MSA-014	TC-MSA-T1-002, TC-MSA-T1-003
UR-MSA-015	TC-MSA-T1-004
UR-MSA-016	TC-MSA-T1-006, TC-MSA-T1-007, TC-MSA-T1-010
UR-MSA-017	TC-MSA-ATT-001, TC-MSA-ATT-002, TC-MSA-ATT-003
UR-MSA-018	TC-MSA-ATT-001, TC-MSA-ATT-004
UR-MSA-019	TC-MSA-ATT-003
UR-MSA-020	TC-MSA-ATT-005
UR-MSA-021	TC-MSA-ATT-008, TC-MSA-ATT-009, TC-MSA-ATT-010
UR-MSA-022	TC-MSA-GRR-010, TC-MSA-NGR-011, TC-MSA-LB-010, TC-MSA-T1-011, TC-MSA-ATT-011
UR-MSA-023	TC-MSA-GRR-005, TC-MSA-NGR-006, TC-MSA-LB-006, TC-MSA-T1-005, TC-MSA-ATT-006
UR-MSA-024	TC-MSA-GRR-006, TC-MSA-NGR-007, TC-MSA-LB-007 (file), TC-MSA-T1-008, TC-MSA-ATT-007

13. Test Cases

Each test case is executed automatically by pytest via repos/msa/admin_msa_oq. Commands shown use wrapper names; the test suite invokes scripts via jrrun.

13.1 jrc_msa_gauge_rr — Gauge R&R (Crossed)

TC-MSA-GRR-001 Happy path — exit 0, output sections present

Tests: UR-MSA-001
Command: `jrc_msa_gauge_rr gauge_rr_balanced.csv`
Pass criterion: Exit code = 0. Output contains: 'ANOVA', 'Variance Components', 'Study Variation', 'Verdict', 'Gauge R&R'.

TC-MSA-GRR-002 Known data — %GRR in expected range, verdict ACCEPTABLE

Tests: UR-MSA-001, UR-MSA-002
Command: `jrc_msa_gauge_rr gauge_rr_balanced.csv`
Rationale: Dataset has low gauge noise; expected %GRR < 10%.
Pass criterion: Exit code = 0. Parsed %GRR < 10%. Verdict = 'ACCEPTABLE'.

TC-MSA-GRR-003 --tolerance flag

Tests: UR-MSA-001
Command: `jrc_msa_gauge_rr gauge_rr_balanced.csv --tolerance 0.5`
Pass criterion: Exit code = 0. Output contains tolerance-referenced %GRR value.

TC-MSA-GRR-004 PNG written to ~/Downloads/

Tests: UR-MSA-003
Command: `jrc_msa_gauge_rr gauge_rr_balanced.csv`
Pass criterion: Exit code = 0. File matching *_jrc_msa_gauge_rr.png exists in ~/Downloads/ with modification time ≥ test start.

TC-MSA-GRR-005 No arguments → usage message

Tests: UR-MSA-023
Command: `jrc_msa_gauge_rr`
Pass criterion: Exit code ≠ 0. Output contains 'Usage' or 'usage'.

TC-MSA-GRR-006 File not found

Tests: UR-MSA-024
Command: `jrc_msa_gauge_rr /tmp/no_such.csv`
Pass criterion: Exit code ≠ 0. Output contains 'not found' (case-insensitive).

TC-MSA-GRR-007 Missing column → column named in error

Tests: UR-MSA-004
Command: `jrc_msa_gauge_rr gauge_rr_missing_col.csv`
Pass criterion: Exit code ≠ 0. Output contains name of missing column.

TC-MSA-GRR-008 Unbalanced design

Tests: UR-MSA-004
Command: `jrc_msa_gauge_rr gauge_rr_unbalanced.csv`
Pass criterion: Exit code ≠ 0. Output contains 'unbalanced' (case-insensitive).

TC-MSA-GRR-009 Single operator

Tests: UR-MSA-004
Command: `jrc_msa_gauge_rr gauge_rr_one_operator.csv`
Pass criterion: Exit code ≠ 0.

TC-MSA-GRR-010 Bypass protection — direct Rscript call

Tests: UR-MSA-022
Command: `Rscript repos/msa/R/jrc_msa_gauge_rr.R gauge_rr_balanced.csv (RENV_PATHS_ROOT unset)`
Rationale: Ensures script cannot be run outside jrrun.
Pass criterion: Exit code ≠ 0. Output contains 'RENV_PATHS_ROOT'.

13.2 jrc_msa_nested_grr — Nested Gauge R&R

TC-MSA-NGR-001 Happy path — exit 0, output sections present

Tests: UR-MSA-005
Command: `jrc_msa_nested_grr nested_grr_good.csv`
Pass criterion: Exit code = 0. Output contains: 'ANOVA', 'Variance Components', 'Study Variation', 'Verdict', 'Gauge R&R', 'Nested'.

TC-MSA-NGR-002 Known good data — %GRR < 30%, verdict ACCEPTABLE or MARGINAL

Tests: UR-MSA-005, UR-MSA-006
Command: `jrc_msa_nested_grr nested_grr_good.csv`
Rationale: Dataset has low gauge noise relative to part variation.
Pass criterion: Exit code = 0. Parsed %GRR < 30%. Verdict = 'ACCEPTABLE' or 'MARGINAL'.

TC-MSA-NGR-003 Known poor data — verdict UNACCEPTABLE

Tests: UR-MSA-006
Command: `jrc_msa_nested_grr nested_grr_poor.csv`
Rationale: Dataset has high within-operator replicate variation.
Pass criterion: Exit code = 0. Verdict = 'UNACCEPTABLE'.

TC-MSA-NGR-004 --tolerance flag

Tests: UR-MSA-005
Command: `jrc_msa_nested_grr nested_grr_good.csv --tolerance 10.0`
Pass criterion: Exit code = 0. Output contains tolerance-referenced output and a percentage value.

TC-MSA-NGR-005 PNG written to ~/Downloads/

Tests: UR-MSA-007
Command: `jrc_msa_nested_grr nested_grr_good.csv`
Pass criterion: Exit code = 0. File matching *_jrc_msa_nested_grr.png exists in ~/Downloads/ with modification time ≥ test start.

TC-MSA-NGR-006 No arguments → usage message

Tests: UR-MSA-023

Command: `jrc_msa_nested_grr`
Pass criterion: Exit code ≠ 0. Output contains 'Usage' or 'usage'.

TC-MSA-NGR-007 File not found

Tests: UR-MSA-024
Command: `jrc_msa_nested_grr /tmp/no_such_nested.csv`
Pass criterion: Exit code ≠ 0. Output contains 'not found' (case-insensitive).

TC-MSA-NGR-008 Missing 'replicate' column

Tests: UR-MSA-008
Command: `jrc_msa_nested_grr nested_grr_missing_col.csv`
Pass criterion: Exit code ≠ 0. Output contains 'replicate' (case-insensitive).

TC-MSA-NGR-009 Single operator

Tests: UR-MSA-008
Command: `jrc_msa_nested_grr nested_grr_one_operator.csv`
Pass criterion: Exit code ≠ 0.

TC-MSA-NGR-010 Unbalanced design — unequal parts per operator

Tests: UR-MSA-008
Command: `jrc_msa_nested_grr nested_grr_unbalanced.csv`
Pass criterion: Exit code ≠ 0. Output contains 'unbalanced', 'parts', or 'equal' (case-insensitive).

TC-MSA-NGR-011 Bypass protection — direct Rscript call

Tests: UR-MSA-022
Command: `Rscript repos/msa/R/jrc_msa_nested_grr.R nested_grr_good.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0. Output contains 'RENV_PATHS_ROOT'.

13.3 jrc_msa_linearity_bias — Linearity and Bias

TC-MSA-LB-001 Happy path — exit 0, output sections present

Tests: UR-MSA-009
Command: `jrc_msa_linearity_bias linearity_bias_good.csv`
Pass criterion: Exit code = 0. Output contains: 'Linearity', 'Bias', 'Slope', 'Intercept', 'Verdict'.

TC-MSA-LB-002 Known data — slope and %Linearity in expected range

Tests: UR-MSA-009, UR-MSA-010
Command: `jrc_msa_linearity_bias linearity_bias_good.csv`
Rationale: Dataset has known linear bias; slope ≈ 0.05.
Pass criterion: Exit code = 0. Parsed slope in [0.01, 0.15]. %Linearity > 0.

TC-MSA-LB-003 --tolerance flag

Tests: UR-MSA-009
Command: `jrc_msa_linearity_bias linearity_bias_good.csv --tolerance 5.0`
Pass criterion: Exit code = 0. Output references tolerance.

TC-MSA-LB-004 Verdict present

Tests: UR-MSA-010
Command: `jrc_msa_linearity_bias linearity_bias_good.csv`
Pass criterion: Exit code = 0. Output contains 'ACCEPTABLE' or 'UNACCEPTABLE'.

TC-MSA-LB-005 PNG written to ~/Downloads/

Tests: UR-MSA-011
Command: `jrc_msa_linearity_bias linearity_bias_good.csv`

Pass criterion: File matching *_jrc_msa_linearity_bias.png exists in ~/Downloads/ with modification time ≥ test start.

TC-MSA-LB-006 No arguments → usage message

Tests: UR-MSA-023
Command: `jrc_msa_linearity_bias`
Pass criterion: Exit code ≠ 0. Output contains 'Usage' or 'usage'.

TC-MSA-LB-007 File not found

Tests: UR-MSA-024
Command: `jrc_msa_linearity_bias /tmp/no_such_lb.csv`
Pass criterion: Exit code ≠ 0. Output contains 'not found' (case-insensitive).

TC-MSA-LB-008 Missing column

Tests: UR-MSA-012
Command: `jrc_msa_linearity_bias linearity_bias_missing_col.csv`
Pass criterion: Exit code ≠ 0. Output names missing column.

TC-MSA-LB-009 Single reference level

Tests: UR-MSA-012
Command: `jrc_msa_linearity_bias linearity_bias_one_part.csv`
Pass criterion: Exit code ≠ 0.

TC-MSA-LB-010 Inconsistent reference per part

Tests: UR-MSA-012
Command: `jrc_msa_linearity_bias linearity_bias_inconsistent_ref.csv`
Pass criterion: Exit code ≠ 0.

TC-MSA-LB-011 Bypass protection — direct Rscript call

Tests: UR-MSA-022
Command: `Rscript repos/msa/R/jrc_msa_linearity_bias.R linearity_bias_good.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0. Output contains 'RENV_PATHS_ROOT'.

13.4 jrc_msa_type1 — Type 1 Gauge Study

TC-MSA-T1-001 Happy path — exit 0, output sections present

Tests: UR-MSA-013
Command: `jrc_msa_type1 type1_good.csv --reference 50.0 --tolerance 1.0`
Pass criterion: Exit code = 0. Output contains: 'Cg', 'Cgk', 'Bias', 'Verdict'.

TC-MSA-T1-002 Known good data — Cg ≥ 1.33, Cgk ≥ 1.33, verdict CAPABLE

Tests: UR-MSA-013, UR-MSA-014
Command: `jrc_msa_type1 type1_good.csv --reference 50.0 --tolerance 1.0`
Rationale: Dataset: 25 measurements near reference, low noise.
Pass criterion: Exit code = 0. Parsed Cg ≥ 1.33. Parsed Cgk ≥ 1.33. Verdict = 'CAPABLE'.

TC-MSA-T1-003 Biased data — Cgk < Cg

Tests: UR-MSA-014
Command: `jrc_msa_type1 type1_biased.csv --reference 50.0 --tolerance 1.0`
Rationale: Dataset has systematic positive bias.
Pass criterion: Exit code = 0. Parsed Cgk < Cg.

TC-MSA-T1-004 PNG written to ~/Downloads/

Tests: UR-MSA-015
Command: `jrc_msa_type1 type1_good.csv --reference 50.0 --tolerance 1.0`

Pass criterion: File matching *_jrc_msa_type1.png exists in ~/Downloads/ with modification time ≥ test start.

TC-MSA-T1-005 No arguments → usage message

Tests: UR-MSA-023
Command: `jrc_msa_type1`
Pass criterion: Exit code ≠ 0. Output contains 'Usage' or 'usage'.

TC-MSA-T1-006 --reference missing

Tests: UR-MSA-016
Command: `jrc_msa_type1 type1_good.csv --tolerance 1.0`
Pass criterion: Exit code ≠ 0. Output contains 'reference' (case-insensitive).

TC-MSA-T1-007 --tolerance missing

Tests: UR-MSA-016
Command: `jrc_msa_type1 type1_good.csv --reference 50.0`
Pass criterion: Exit code ≠ 0. Output contains 'tolerance' (case-insensitive).

TC-MSA-T1-008 File not found

Tests: UR-MSA-024
Command: `jrc_msa_type1 /tmp/no_such_t1.csv --reference 50.0 --tolerance 1.0`
Pass criterion: Exit code ≠ 0. Output contains 'not found' (case-insensitive).

TC-MSA-T1-009 Missing 'value' column

Tests: UR-MSA-016
Command: `jrc_msa_type1 type1_missing_value_col.csv --reference 50.0 --tolerance 1.0`
Pass criterion: Exit code ≠ 0. Output contains 'value' (case-insensitive).

TC-MSA-T1-010 Too few measurements (< 10)

Tests: UR-MSA-016
Command: `jrc_msa_type1 type1_too_few.csv --reference 50.0 --tolerance 1.0`
Pass criterion: Exit code ≠ 0.

TC-MSA-T1-011 Bypass protection — direct Rscript call

Tests: UR-MSA-022
Command: `Rscript repos/msa/R/jrc_msa_type1.R type1_good.csv --reference 50.0 --tolerance 1.0  (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0. Output contains 'RENV_PATHS_ROOT'.

13.5 jrc_msa_attribute — Attribute Agreement Analysis

TC-MSA-ATT-001 Dataset with reference → exit 0, all sections present

Tests: UR-MSA-017, UR-MSA-018
Command: `jrc_msa_attribute attribute_with_ref.csv`
Pass criterion: Exit code = 0. Output contains: 'Within-Appraiser', 'Between-Appraiser', 'vs Reference' (or 'Appraiser vs'), 'Verdict', 'Fleiss'.

TC-MSA-ATT-002 Dataset without reference → exit 0, vs-reference section absent

Tests: UR-MSA-017
Command: `jrc_msa_attribute attribute_no_ref.csv`
Pass criterion: Exit code = 0. Output contains 'Within-Appraiser' and 'Between-Appraiser'. Output does NOT contain 'Vs Reference'.

TC-MSA-ATT-003 Known data — Fleiss' Kappa in expected range, verdict present

Tests: UR-MSA-017, UR-MSA-019

Command: `jrc_msa_attribute attribute_with_ref.csv`
Rationale: Known dataset; Fleiss' Kappa expected in [0.7, 1.0].
Pass criterion: Exit code = 0. Parsed Kappa ∈ [0.7, 1.0]. Verdict = 'MARGINAL' or 'ACCEPTABLE'.

TC-MSA-ATT-004 Perfect appraiser vs reference — Kappa = 1.0

Tests: UR-MSA-018
Command: `jrc_msa_attribute attribute_with_ref.csv`
Rationale: Appraiser A agrees with reference on all trials in the test dataset.
Pass criterion: Exit code = 0. Parsed Kappa for appraiser A vs reference = 1.0000.

TC-MSA-ATT-005 PNG written to ~/Downloads/

Tests: UR-MSA-020
Command: `jrc_msa_attribute attribute_with_ref.csv`
Pass criterion: File matching *_jrc_msa_attribute.png exists in ~/Downloads/ with modification time ≥ test start.

TC-MSA-ATT-006 No arguments → usage message

Tests: UR-MSA-023
Command: `jrc_msa_attribute`
Pass criterion: Exit code ≠ 0. Output contains 'Usage' or 'usage'.

TC-MSA-ATT-007 File not found

Tests: UR-MSA-024
Command: `jrc_msa_attribute /tmp/no_such_attr.csv`
Pass criterion: Exit code ≠ 0. Output contains 'not found' (case-insensitive).

TC-MSA-ATT-008 Missing 'trial' column

Tests: UR-MSA-021
Command: `jrc_msa_attribute attribute_missing_col.csv`
Pass criterion: Exit code ≠ 0. Output contains 'trial' (case-insensitive).

TC-MSA-ATT-009 Only one appraiser

Tests: UR-MSA-021
Command: `jrc_msa_attribute attribute_one_appraiser.csv`
Pass criterion: Exit code ≠ 0.

TC-MSA-ATT-010 Unbalanced design

Tests: UR-MSA-021
Command: `jrc_msa_attribute attribute_unbalanced.csv`
Pass criterion: Exit code ≠ 0. Output contains 'unbalanced', 'trials', or 'equal' (case-insensitive).

TC-MSA-ATT-011 Bypass protection — direct Rscript call

Tests: UR-MSA-022
Command: `Rscript repos/msa/R/jrc_msa_attribute.R attribute_with_ref.csv (RENV_PATHS_ROOT unset)`
Pass criterion: Exit code ≠ 0. Output contains 'RENV_PATHS_ROOT'.

14. Deviation Procedure

Any test case that does not meet its pass criterion constitutes a deviation. Deviations shall be handled as follows:

14. Record the deviation in the OQ execution evidence, noting: test case ID, actual result, and expected result
15. Investigate the root cause (software defect, test data error, environmental issue)

16. If a software defect is identified: correct the script, regenerate the integrity file, and re-execute the full test suite
17. If a test data error is identified: correct the test data, document the change, and re-execute the affected test cases
18. Document the resolution and obtain QA approval before releasing the module

The OQ shall not be considered successful until all deviations are resolved and all 53 test cases pass.